A12 AIR FILTRATION | O (MAX : 1 PT)

Intent : Reduce indoor and outdoor airborne contaminants through air filtration.

Summary : This WELL feature requires projects with mechanically ventilated spaces to implement adequate air filtration and document a maintenance protocol for installed filters.

Issue : Exposure to particulate matter (PM) is associated with many negative health outcomes. PM_{10} can block and inflame airways, causing a range of respiratory-related conditions that can lead to illness or death.[1] $PM_{2.5}$ poses even greater health risks compared to PM_{10}, because it can penetrate deep into the lungs, enter the bloodstream and as a result, cause a variety of health issues, including heart disease and other cardiovascular complications.[1]

Solutions : Selection and installation of adequate media filters is one of the key mechanisms for minimizing exposure to outdoor and indoor air pollution. Studies have shown that decreased exposure to particulate matter by filtration of recirculated indoor air is associated with a reduced risk of cardiovascular disease and is an effective control measure for reducing allergic respiratory disease.[2,3] In addition, regular filter maintenance is critical to ensure proper air filtration and the efficiency of the air conditioning system. During the operation, filters should be replaced when they become loaded with particles, since they will begin to reduce airflow and increase pressure drop. Overloaded filters not only restrict airflow rate but can also result in a loss of filtration efficiency. It is also critical for projects to be aware whether the building is located in an area with elevated outdoor air pollution, since these projects often need to install a pre-filtration stage, in addition to the primary filtration, to maintain high-quality indoor air.[4]

PART 1 IMPLEMENT PARTICLE FILTRATION (MAX : 1 PT)

For All Spaces:

1: Filtration levels

The following requirement is met:

a. Media filters are used in the ventilation system to filter outdoor air supplied to the space, in accordance with thresholds specified in the table below:[5,6]

Annual Average Outdoor $PM_{2.5}$ Threshold	Average Air Filtration Efficiency (particles 0.3-1 µm)
23 µg/m^3 or less	≥ 35% (e.g., MERV 12 or M6)
24–39 µg/m^3	≥ 75% (e.g., MERV 14, F8 or ePM1 75%)
40 µg/m^3 or greater	≥ 95% (e.g., MERV 16, E10 or ePM1 95%)

2: Filter maintenance

The following requirement is met:

a. Evidence that the filter has been replaced according to the manufacturer's recommendation is submitted annually through the WELL digital platform.

Note :
The World Health Organization's Global Urban Ambient Air Pollution Database may be consulted to view outdoor air quality levels, available at https://www.who.int/data/gho/data/themes/air-pollution/who-air-quality-database

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